

Forecasting U.S. Income Inequality with Autoregressive(AR) Models

A Layered Approach Using the Gini Coefficient

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Plan

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- 3 Layer 1: Linear Trend
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- 5 Layer 3: Extended Models
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- 7 Conclusion

Introduction

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- Income inequality in the U.S. has been rising for decades
- Economists are great at describing what has already happened — but can we actually predict where it is going?

That is exactly what math is for.

The Gini Coefficient

- Measures income inequality in a population

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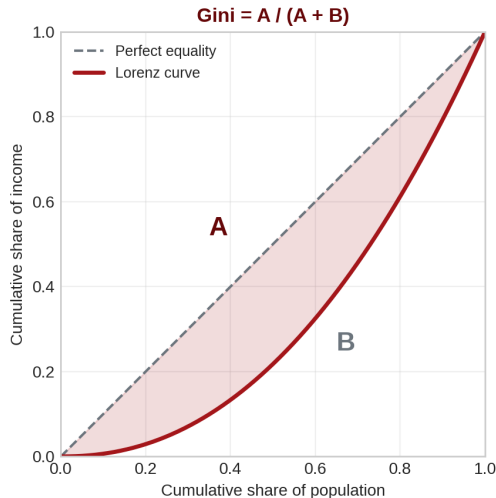
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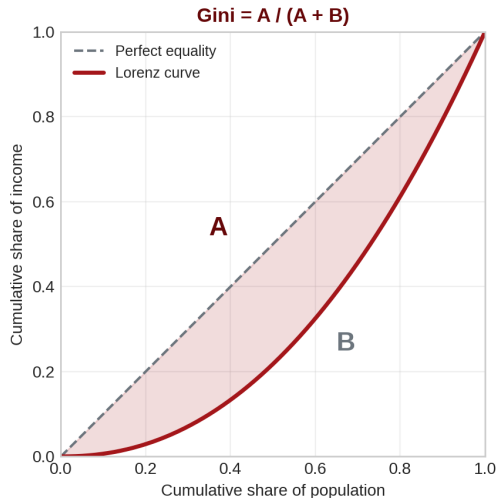


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- A : area between the equality line and the curve
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- The bigger A is relative to B , the more unequal the distribution



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3	AR Families	Adjusts for trends, past mistakes, and economic data

Data

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- Exogenous variables (Layer 3):
 - GDP (Gross Domestic Product) growth
 - CPI (Consumer Price Index) inflation
 - Unemployment rate



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Expanding-window rolling forecast (no look-ahead):

- 1 Fit model on all data from 1963 to year $t-1$
- 2 Predict one step ahead: $\hat{y}_t$
- 3 Reveal actual y_t , expand training window
- 4 Repeat for $t = 1963, 1964, \dots, 1994$

How we measure forecast quality:

$$\text{MAE (Mean Absolute Error)} = \frac{1}{n} \sum_{t=1}^n |e_t|$$

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Results (30 test years, 1994–2023):

MAE	0.0081
RMSE	0.0106
MAPE	2.01%

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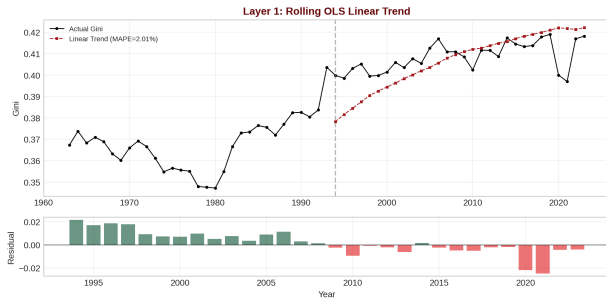
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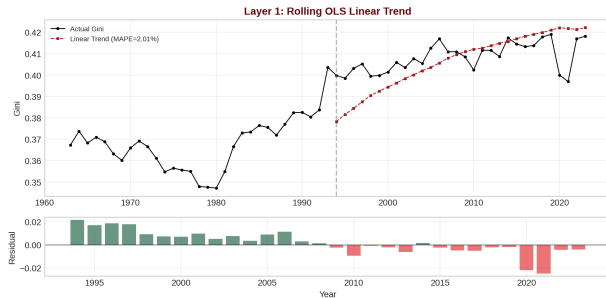
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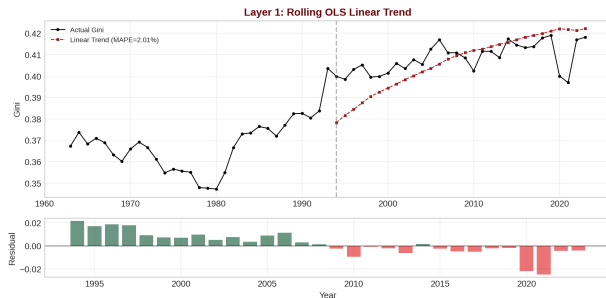
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- Trend over-predicts in the 1990s–2000s
- Residuals show a **pattern** — not random

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Example (5 selected test years):

Year	Actual G_t	Predicted $\hat{G}_t$	Error	$\left \frac{e_t}{G_t} \right \times 100$
1999	0.3998	0.3925	0.0073	1.83%
2003	0.4077	0.4001	0.0076	1.86%
2005	0.4126	0.4036	0.0090	2.18%
2010	0.4024	0.4121	0.0097	2.40%
2020	0.4000	0.4221	0.0221	5.52%
Average of all 30 test years:				2.01%

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- Most years: error < 2.5%
- 2020 COVID spike pulls the average up
- MAPE < 5% $\Rightarrow$ generally considered highly accurate

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Errors follow a pattern $\longrightarrow$ we need a model that **can learn from its own past values**

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How many past years should we use?

Too few lags

Misses real patterns
(underfitting)

VS

Just right

AIC / BIC pick this
Lowest score wins

VS

Too many lags

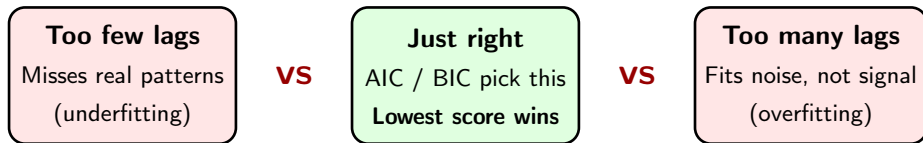
Fits noise, not signal
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Autoregressive (AR) model:

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How many past years should we use?



- AIC and BIC score each candidate model using its **likelihood** (how well it fits) minus a **penalty** (how many parameters it uses)

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- ϕ_i captures **persistence**: how strongly the past affects the present
- ε_t are errors
- But how many lags p should we use?

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p=6	-180.09	-170.34
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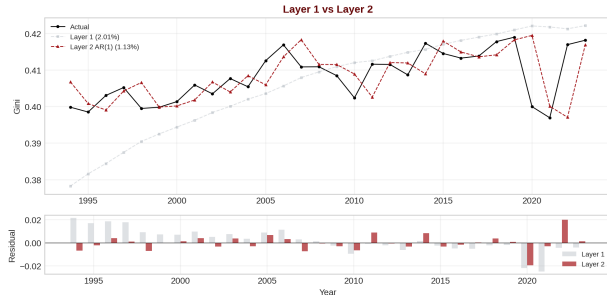
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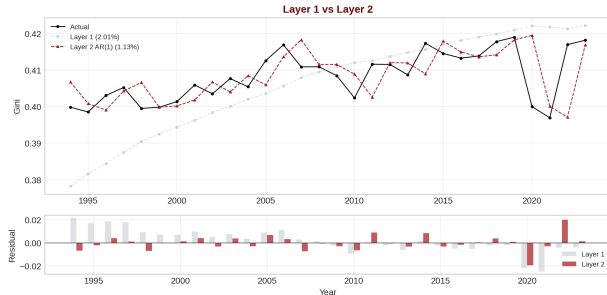
	L1	L2	Improvement
MAE	.0081	.0046	43%
RMSE	.0106	.0066	38%
MAPE	2.01%	1.13%	44%

Layer 2: Forecast vs. Actual



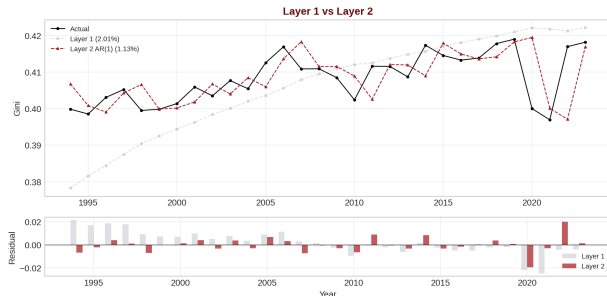
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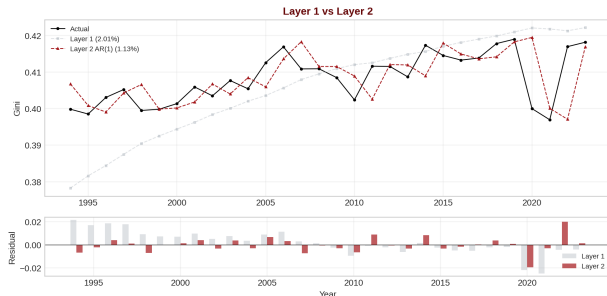
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- Layer 2 tracks the actual data much more closely

Layer 2: Forecast vs. Actual



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- Biggest miss: 2020 COVID drop (-0.020)

Layer 1 vs. Layer 2: Error by Year

Year	Actual y_t	L1 error	L1 $\left \frac{e}{y} \right $	L2 error	L2 $\left \frac{e}{y} \right $
1999	0.3998	+0.0073	1.83%	-0.0000	0.01%
2005	0.4126	+0.0090	2.18%	+0.0066	1.60%
2010	0.4024	-0.0097	2.40%	-0.0065	1.62%
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All 30 years:			2.01%		1.13%

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- **2020:** COVID caused a sudden drop that **neither model can anticipate** — this is what motivates Layer 3 (adding economic data)

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We need a model that can use **economic data**, **correct its own errors**, and handle **trending data** → **Layer 3**

Layer 3: Extended Models

Layer 3: Extending AR(1)

Layer 3 tests four extensions, each adding **one capability**:

Model	Past Gini	Past errors	Differenced	Macro data
AR(1)	✓			
ARX	✓			✓
ARMA	✓	✓		
ARIMA	✓	✓	✓	
ARIMAX	✓	✓	✓	✓

3A: ARX — What Does “X” Mean?

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Our macro regressors (from FRED):

- ΔGDP_t : year-over-year change in GDP
- ΔCPI_t : change in inflation
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General ARX model:

$$y_t = c + \phi_1 y_{t-1} + \sum_k \beta_k \Delta X_{t-k} + \varepsilon_t$$

3A: ARX — BIC Selection

Model	AIC	BIC
AR(1) + dGDP_lag2 + dCPI_lag1	-208.39	-199.91 <<<
AR(1) + dGDP_lag2 + dCPI_lag1 + dCPI_lag3	-209.02	-199.24
AR(1) + dGDP_lag2 + dCPI_lag1 + dCPI_lag2	-208.72	-198.94
AR(1) + dCPI_lag2	-205.78	-198.59
AR(1) + dGDP_lag1 + dGDP_lag2 + dCPI_lag1	-207.35	-197.57
AR(1) + dGDP_lag2 + dCPI_lag2	-205.97	-197.49
AR(1) + dCPI_lag1	-204.50	-197.32
AR(1) + dGDP_lag1 + dGDP_lag2 + dCPI_lag2	-207.01	-197.23
AR(1) + dGDP_lag2 + dCPI_lag1 + dUNRATE_lag2	-206.89	-197.11
AR(1) + dGDP_lag1 + dGDP_lag2 + dCPI_lag1 + dCPI_lag2	-208.11	-197.04

>>> BIC selects: AR(1) + dGDP_lag2 + dCPI_lag1 (BIC = -199.9112)

3A: ARX — Model Equation

Selected specification:

$$y_t = c + \phi_1 y_{t-1} + \beta_1 \Delta \text{GDP}_{t-2} + \beta_2 \Delta \text{CPI}_{t-1} + \varepsilon_t$$

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Worked example: predicting y_{1994}

Training on 1963–1993 gives:

$$\hat{c} = -0.1236, \quad \hat{\phi}_1 = 1.3319, \quad \hat{\beta}_1 = -0.00004, \quad \hat{\beta}_2 = 0.0027$$

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$$\begin{aligned} \hat{y}_{1994} &= -0.1236 + 1.3319 \times \underbrace{0.4036}_{y_{1993}} + (-0.00004) \times \underbrace{362.2}_{\Delta \text{GDP}_{1992}} + 0.0027 \times \underbrace{4.17}_{\Delta \text{CPI}_{1993}} \\ &= 0.4120 \end{aligned}$$

Actual $y_{1994} = 0.3999$, error = $0.3999 - 0.4120 = -0.0121$

3A: ARX — Forecast vs. Actual



3A: ARX — Results & Comparison

Model	MAE	RMSE	MAPE (%)
Layer 2: AR(1)	0.0046	0.0066	1.13
3A: ARX	0.0048	0.0059	1.17

Takeaway: Macro variables do **not** improve forecasts beyond lagged Gini alone. The ΔGDP coefficient is essentially zero ($\hat{\beta}_1 \approx -0.00004$). Only ΔCPI has a nonzero coefficient, but it does not help out-of-sample.

Interlude: From OLS to Maximum Likelihood

Starting from ARMA, we need a more powerful estimation method called **conditional maximum likelihood estimation** (MLE).

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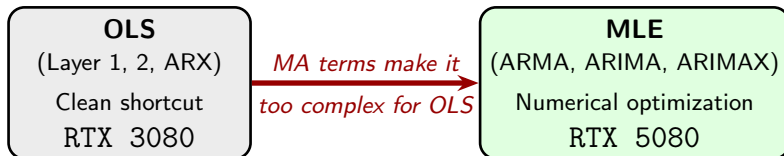
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Same goal, same game — just more horsepower because the MA terms make the problem too complex for the old card.

We still use **BIC** for model selection.

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General ARMA(p, q) model:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

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BIC selection over ARMA(p, q), $p, q \in \{0, 1, 2, 3\}$:

3B: ARMA — BIC Selection

(p,q)	AIC	BIC
(0,1)	-184.08	-178.09
(0,2)	-217.84	-210.51
(0,3)	-203.39	-194.73
(1,0)	-206.35	-200.35
(1,1)	-208.58	-201.25
(1,2)	-207.80	-199.14
(1,3)	-209.59	-199.60
(2,0)	-207.90	-200.57
(2,1)	-207.09	-198.43
(2,2)	-248.45	-238.46
(2,3)	-230.10	-218.77
(3,0)	-207.69	-199.03
(3,1)	-206.57	-196.57
(3,2)	-207.16	-195.84
(3,3)	-216.12	-203.46

>>> BIC selects ARMA(2,2) (BIC = -238.4553)

3B: ARMA(2,2) — Model Equation

Selected specification:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t$$

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$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t$$

Worked example: predicting y_{1994}

MLE on 1963–1993 gives:

$$\hat{c} = -0.121, \quad \hat{\phi}_1 = 0.689, \quad \hat{\phi}_2 = 0.629, \quad \hat{\theta}_1 = 2.858, \quad \hat{\theta}_2 = 2.405$$

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The forecast uses past residuals $\varepsilon_{t-1}, \varepsilon_{t-2}$ computed internally by MLE:

$$\hat{y}_{1994} = 0.4198$$

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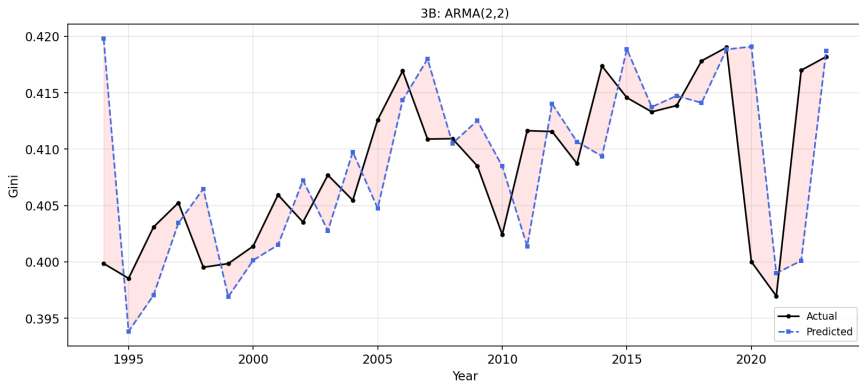
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$$\hat{y}_{1994} = 0.4198$$

Actual $y_{1994} = 0.3999$, error = -0.0200

3B: ARMA(2,2) — Forecast vs. Actual



3B: ARMA(2,2) — Results & Comparison

Model	MAE	RMSE	MAPE (%)
Layer 2: AR(1)	0.0046	0.0066	1.13
3B: ARMA(2,2)	0.0053	0.0074	1.31

Takeaway: Error correction adds complexity but does **not** improve out-of-sample accuracy. The MA terms overfit to training-period patterns.

3C: ARIMA — What Does “I” Mean?

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The problem: $\hat{\phi}_1 \approx 0.97$ means every Gini value is almost the same number — they're all crowded between 0.39 and 0.42. It's hard to find real patterns when all your data points look nearly identical.

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The fix: Model $\Delta y_t = y_t - y_{t-1}$ instead. The changes bounce around zero — some positive, some negative — giving the estimation something meaningful to work with.

3C: ARIMA — BIC Selection

ARIMA(p, d, q): difference d times, then fit ARMA(p, q) on Δy_t .

General ARIMA($p, 1, q$) model:

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \cdots + \phi_p \Delta y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

We fix $d = 1$ and search $p, q \in \{0, 1, 2, 3\}$:

($p, 1, q$)	AIC	BIC
(0, 1, 1)	-203.04	-197.15
(0, 1, 2)	-203.55	-196.37
(0, 1, 3)	-202.22	-193.75
(1, 1, 0)	-201.91	-196.03
(1, 1, 1)	-201.44	-194.26
(1, 1, 2)	-201.82	-193.34
(1, 1, 3)	-200.30	-190.53
(2, 1, 0)	-202.46	-195.28
(2, 1, 1)	-200.56	-192.08
(2, 1, 2)	-200.24	-190.46
(2, 1, 3)	-198.82	-187.75
(3, 1, 0)	-200.49	-192.01
(3, 1, 1)	-199.55	-189.78
(3, 1, 2)	-198.85	-187.78
(3, 1, 3)	-222.62	-210.25

3C: ARIMA(3,1,3) — Model Equation

Selected specification:

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \phi_2 \Delta y_{t-2} + \phi_3 \Delta y_{t-3} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + \varepsilon_t$$

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MLE on 1963–1993 gives:

$$\hat{c} = -0.001, \quad \hat{\phi}_1 = -1.637, \quad \hat{\phi}_2 = -1.716, \quad \hat{\phi}_3 = -0.207, \quad \hat{\theta}_1 = 0.786, \quad \hat{\theta}_2 = 3.378, \quad \hat{\theta}_3 = 2.351$$

3C: ARIMA(3,1,3) — Worked Example

Predicting y_{1994} :

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Predicting y_{1994} :

Step 1. Difference: $\Delta y_{1993} = 0.0199$, $\Delta y_{1992} = 0.0033$, $\Delta y_{1991} = -0.0022$

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Step 2. MLE also computes past residuals internally:

$$\varepsilon_{1993} = 0.0020, \varepsilon_{1992} = -0.0001, \varepsilon_{1991} = 0.0059$$

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Step 3. Predict the change:

$$\begin{aligned}\widehat{\Delta y}_{1994} &= \underbrace{-0.001}_c + \underbrace{(-1.637)(0.0199)}_{\phi_1 \Delta y_{1993}} + \underbrace{(-1.716)(0.0033)}_{\phi_2 \Delta y_{1992}} + \underbrace{(-0.207)(-0.0022)}_{\phi_3 \Delta y_{1991}} \\ &\quad + \underbrace{(0.786)(0.0020)}_{\theta_1 \varepsilon_{1993}} + \underbrace{(3.378)(-0.0001)}_{\theta_2 \varepsilon_{1992}} + \underbrace{(2.351)(0.0059)}_{\theta_3 \varepsilon_{1991}} \\ &= -0.0235\end{aligned}$$

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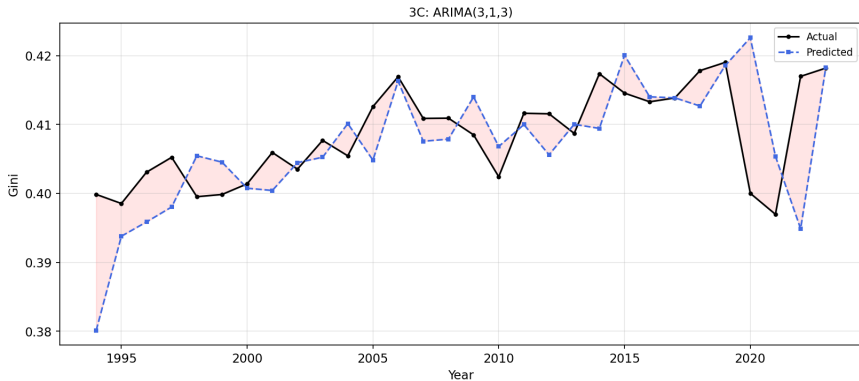
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Step 4. Convert back: $\hat{y}_{1994} = y_{1993} + \widehat{\Delta y}_{1994} = 0.4036 + (-0.0235) = 0.3801$

Actual $y_{1994} = 0.3999$, error = +0.0197

3C: ARIMA(3,1,3) — Forecast vs. Actual



3C: ARIMA(3,1,3) — Results & Comparison

Model	MAE	RMSE	MAPE (%)
Layer 2: AR(1)	0.0046	0.0066	1.13
3C: ARIMA(3,1,3)	0.0057	0.0082	1.40

Takeaway: Differencing + MA does not beat AR(1). The high-order model (6 extra parameters) overfits the small 31-year training set.

3D: ARIMAX — ARIMA + Exogenous Variables

ARIMAX combines everything: differencing, MA correction, and macro regressors.

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \cdots + \beta_k \Delta X_{t-\ell_k} + \cdots + \theta_1 \varepsilon_{t-1} + \cdots + \varepsilon_t$$

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BIC subset selection on exogenous variables:

Using ARIMA(3,1,3) as baseline, we test all macro variable combinations.

3D: ARIMAX — BIC Selection

```
Using ARIMA(3,1,3) from ARIMA selection as baseline  
Baseline BIC (no exog) = -210.2512
```

Exogenous variables	BIC
No exog (ARIMA only)	-210.25 <<<

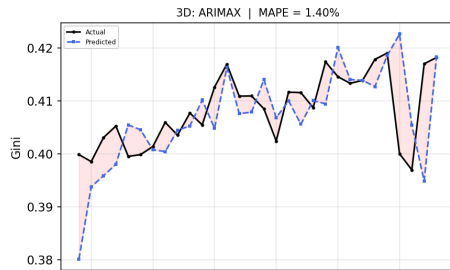
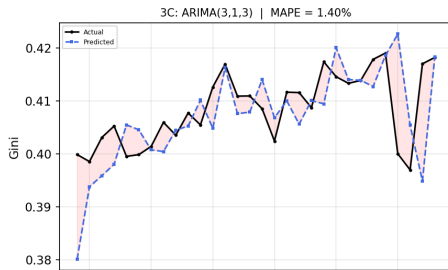
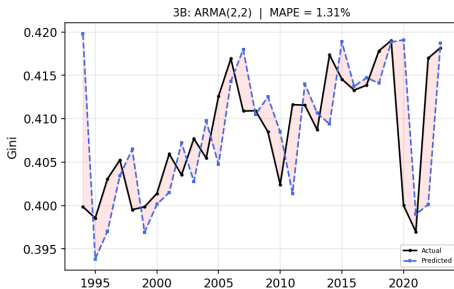
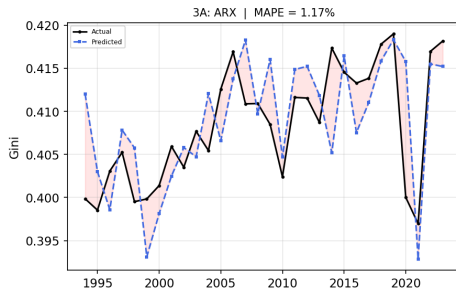
```
>>> No exogenous variables improve BIC – ARIMAX falls back to ARIMA
```

ARIMAX = ARIMA(3,1,3) — macro data does **not** help on top of ARIMA.

Results

Model Comparison: All Four Extensions

Layer 3: All Models vs Actual (1994-2023)



Model Comparison: Metrics

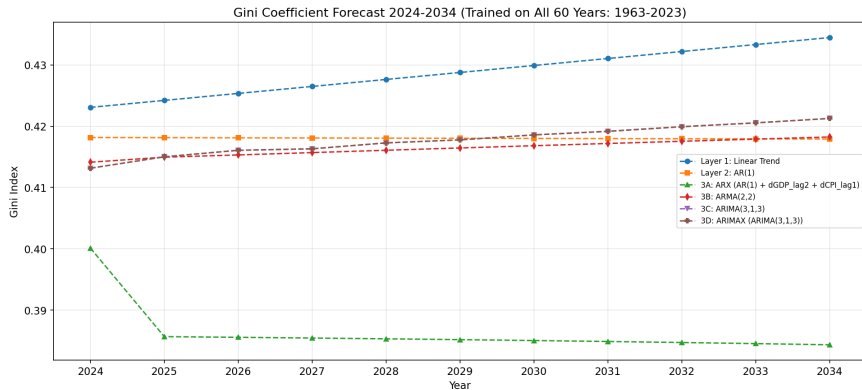
Model	MAE	RMSE	MAPE (%)
Layer 1: Linear Trend	0.0081	0.0106	2.01
Layer 2: AR(1)	0.0046	0.0066	1.13
3A: ARX	0.0048	0.0059	1.17
3B: ARMA(2,2)	0.0053	0.0074	1.31
3C: ARIMA(3,1,3)	0.0057	0.0082	1.40
3D: ARIMAX	0.0057	0.0082	1.40

Key takeaway: AR(1) remains the best forecaster.

Adding complexity does **not** improve out-of-sample prediction. Inequality is dominated by its own persistence.

Forecasting: 2024–2034

Using **all 61 years** (1963–2023) as training data, we forecast forward with all six models.



Conclusion

Limitations:

- Only 61 annual observations — small for complex time-series models
- The Gini compresses an entire income distribution into **one number**
 - Misses top-1% dynamics, wealth vs. income inequality

Future directions:

- **Alternative measures:** Palma ratio, top income shares, wealth Gini
- **Higher-frequency data:** quarterly or monthly inequality proxies

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Questions?